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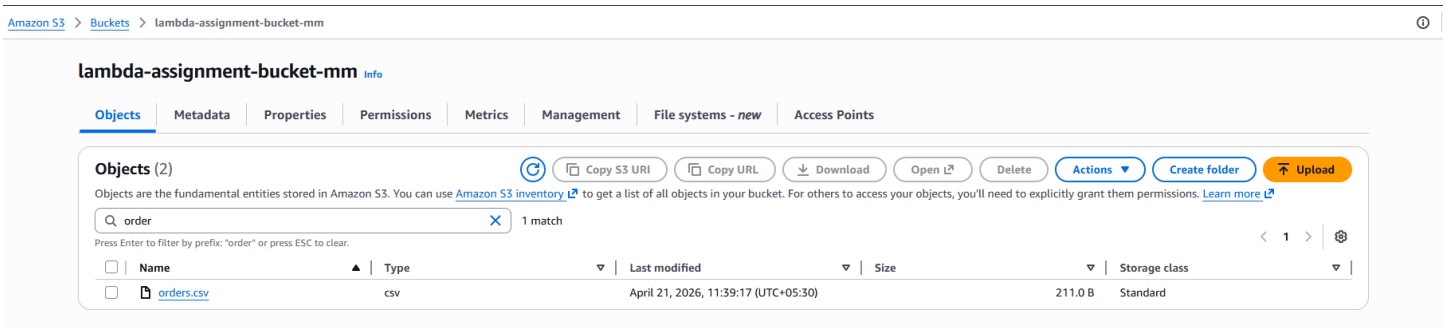
AWS Lambda Assignment

Task 1 : Cron-Based

• Use a Lambda function to aggregate the number of completed, pending and failed orders for each user

Bucket Configurations :-



Results after Testing :-

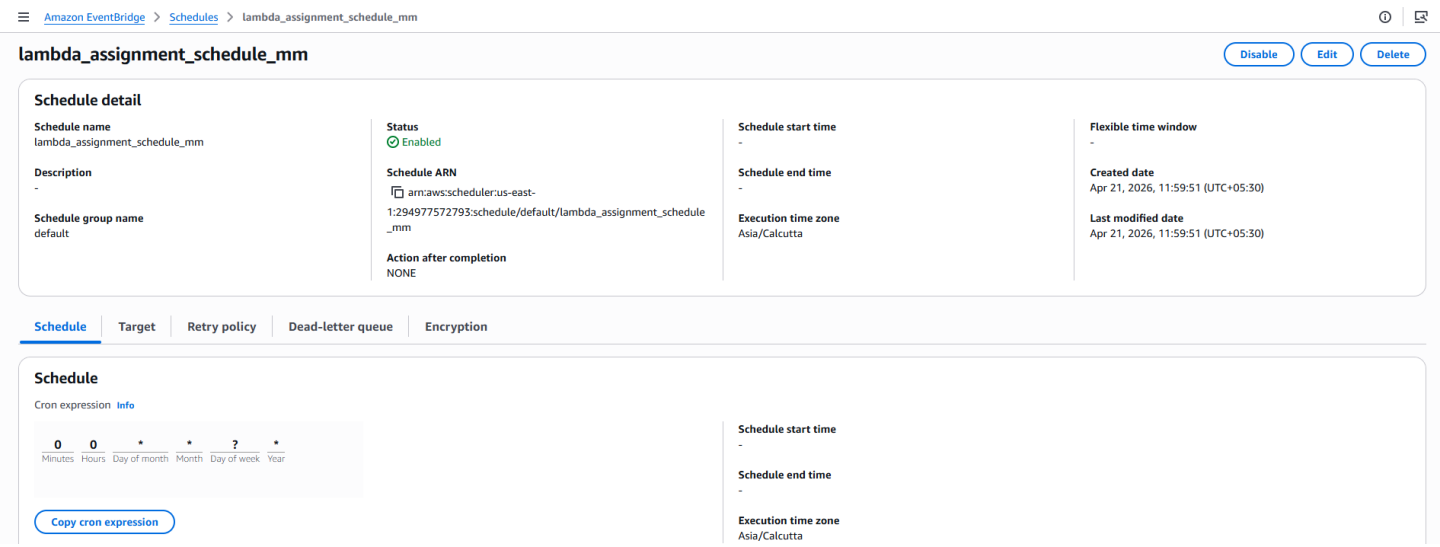
Script of Lambda Function:-

```python
import json
import boto3
import csv
from collections import defaultdict

s3 = boto3.client('s3')

BUCKET_NAME = "lambda-assignment-bucket-mm"
FILE_KEY = "orders.csv"

def lambda_handler(event, context):

    response = s3.get_object(Bucket=BUCKET_NAME, Key=FILE_KEY)
    content = response['Body'].read().decode('utf-8-sig').splitlines()

    reader = csv.DictReader(content)

    reader.fieldnames = [field.strip() for field in reader.fieldnames]

    user_stats = defaultdict(lambda: {
        "Completed": 0,
        "Pending": 0,
        "Failed": 0
    })

    for row in reader:
        row = {k.strip().lower(): v.strip() for k, v in row.items()}

        user_id = row.get("userid")
        status = row.get("status")

        if not user_id or not status:
            continue

        status = status.capitalize()

        if status in user_stats[user_id]:
            user_stats[user_id][status] += 1

    result = dict(user_stats)

    print("Aggregated Result:")
    print(json.dumps(result, indent=2))
```

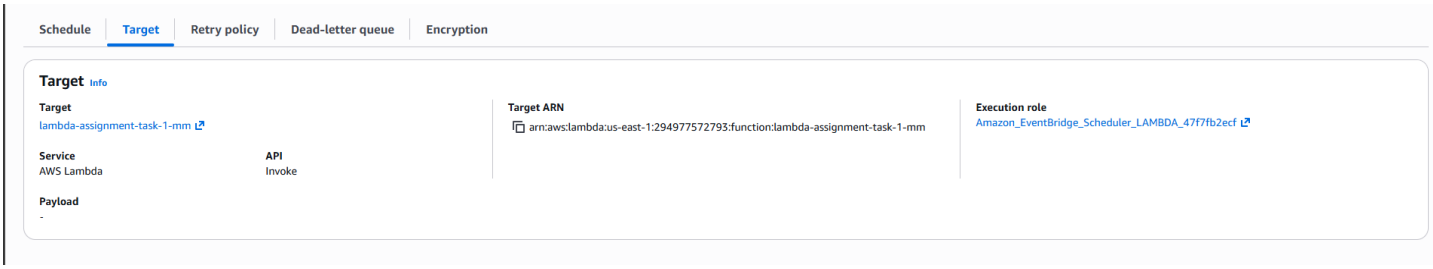
```
    s3.put_object(
        Bucket=BUCKET_NAME,
        Key="aggregated/output.json",
        Body=json.dumps(result),
        ContentType="application/json"
    )

    return {
        "statusCode": 200,
        "body": result
    }
```

- Trigger a Lambda function on daily schedule (e.g., every day at midnight)

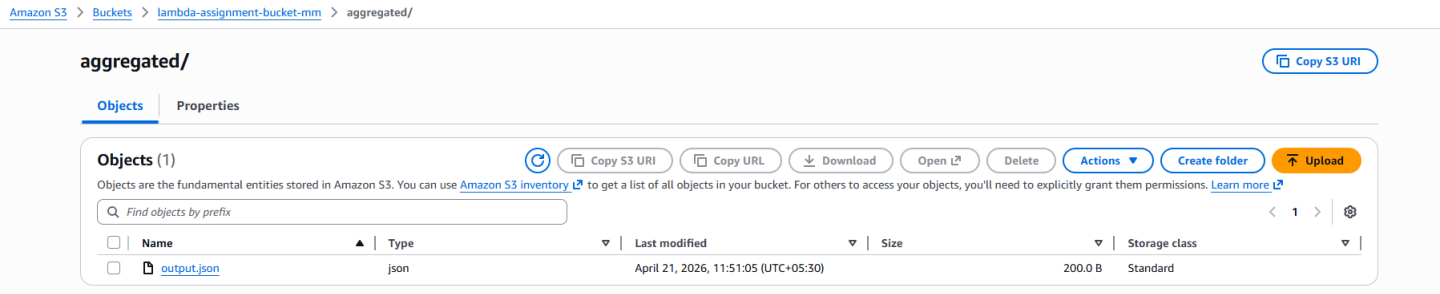
EventBridge Schedule Configurations :-



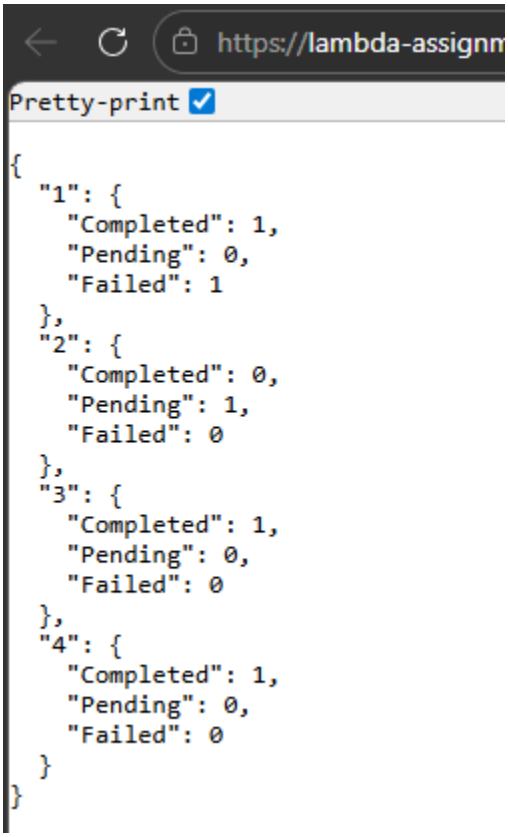
Displaying Target of Schedule :-



Final Output Bucket :-



Final Results :-

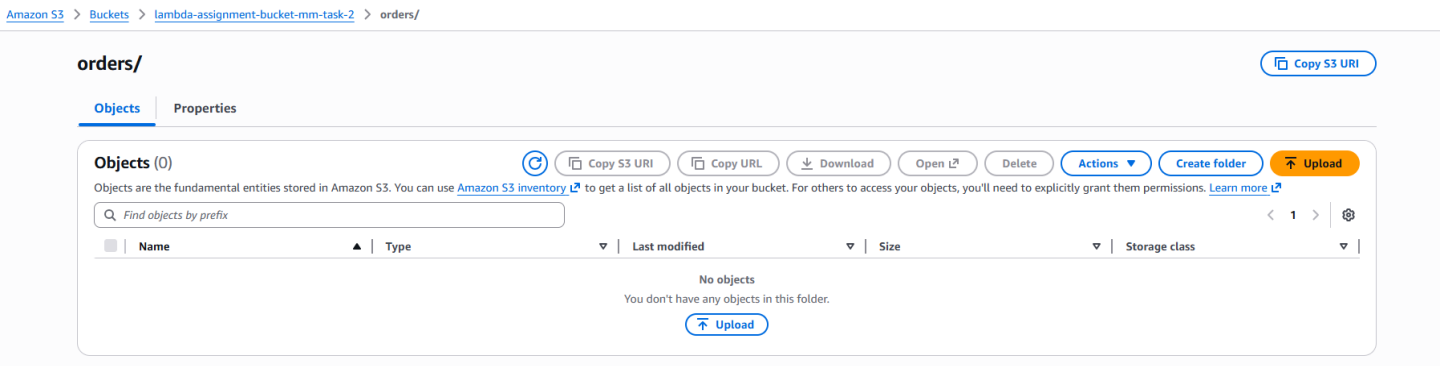



```
{
  "1": {
    "Completed": 1,
    "Pending": 0,
    "Failed": 1
  },
  "2": {
    "Completed": 0,
    "Pending": 1,
    "Failed": 0
  },
  "3": {
    "Completed": 1,
    "Pending": 0,
    "Failed": 0
  },
  "4": {
    "Completed": 1,
    "Pending": 0,
    "Failed": 0
  }
}
```

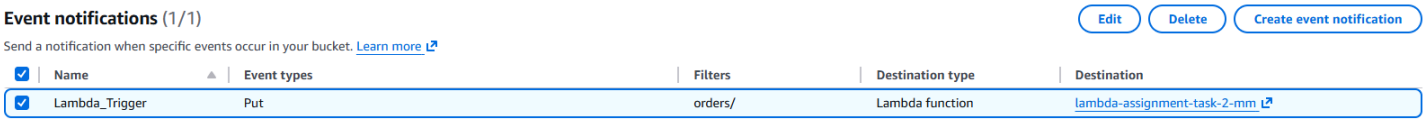
Task 2 : Event-Based Task

- Use an S3 event as the trigger when a new file is uploaded to a specific S3 bucket, a lambda function should process the file.
- Reads the file to check which orders are marked as "Completed".

Initially Empty Orders Folder in S3 :-



Attached Event Notifications for Lambda Function:-



Script :-

```
import json
import boto3
import csv
import urllib.parse
import io

s3 = boto3.client('s3')

def lambda_handler(event, context):

    bucket_name = event['Records'][0]['s3']['bucket']['name']
    file_key = urllib.parse.unquote_plus(event['Records'][0]['s3']['object']['key'])

    print(f"Processing file: {file_key}")

    if not file_key.startswith("orders/"):
        print("Skipping file (not in orders folder)")
        return
```

```python
response = s3.get_object(Bucket=bucket_name, Key=file_key)
content = response['Body'].read().decode('utf-8-sig').splitlines()

reader = csv.DictReader(content)

reader.fieldnames = [field.strip() for field in reader.fieldnames]

completed_orders = []

for row in reader:
    row = {k.strip(): v.strip() for k, v in row.items()}

    status = row.get("Status")

    if status and status.lower() == "completed":
        completed_orders.append(row)

output_buffer = io.StringIO()

if completed_orders:
    fieldnames = completed_orders[0].keys()
    writer = csv.DictWriter(output_buffer, fieldnames=fieldnames)
    writer.writeheader()
    writer.writerows(completed_orders)

csv_data = output_buffer.getvalue()

output_key = file_key.replace("orders/", "processed/")

s3.put_object(
    Bucket=bucket_name,
    Key=output_key,
    Body=csv_data,
    ContentType="text/csv"
)

print(f"Processed file saved to: {output_key}")

return {
    "statusCode": 200,
    "output_file": output_key,
    "records_written": len(completed_orders)
}
```

Uploaded Orders.csv file :-

Amazon S3 > Buckets > lambda-assignment-bucket-mm-task-2 > orders/

orders/

Copy S3 URI

Objects | Properties

Objects (1)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

Find objects by prefix

Name	Type	Last modified	Size	Storage class
orders.csv	csv	April 21, 2026, 14:23:26 (UTC+05:30)	211.0 B	Standard

Logs Generated :-

Log events

Actions | Start tailing | Create metric filter

You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns

Filter events - press enter to search

Clear | 1m | 30m | 1h | 12h | Custom | UTC timezone | Display

Timestamp	Message
	No older events at this moment. *Retry*
2026-04-21T08:53:27.120Z	INIT_START Runtime Version: python:3.14.v39 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:06dae9ab05ccfc05b6ef62a6c3f8c3976f2962ec872732763a914c0041151809
2026-04-21T08:53:27.647Z	START RequestId: e5ac55df-df9a-4189-a1bf-0c2c1b574aea Version: $LATEST
2026-04-21T08:53:27.648Z	Processing file: orders/orders.csv
2026-04-21T08:53:28.011Z	Processed file saved to: processed/orders.csv
2026-04-21T08:53:28.018Z	END RequestId: e5ac55df-df9a-4189-a1bf-0c2c1b574aea
2026-04-21T08:53:28.018Z	REPORT RequestId: e5ac55df-df9a-4189-a1bf-0c2c1b574aea Duration: 370.63 ms Billed Duration: 897 ms Memory Size: 128 MB Max Memory Used: 96 MB Init Duration: 526.13 ms
	No newer events at this moment. *Auto retry paused. Resume*

Processed Orders.csv file in Output :-

Amazon S3 > Buckets > lambda-assignment-bucket-mm-task-2 > processed/

processed/

Copy S3 URI

Objects | Properties

Objects (1)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

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Find objects by prefix

Name	Type	Last modified	Size	Storage class
orders.csv	csv	April 21, 2026, 14:23:28 (UTC+05:30)	142.0 B	Standard

orders.csv

C: > Users > mihir.mungara > Downloads > orders.csv

```
OrderID,UserID,OrderDate,Amount,Status
101,1,2024-05-01,50.00,Completed
103,3,2024-05-03,50.00,Completed
105,4,2024-05-05,50.00,Completed
```